

Metric Spaces

Second Edition

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Preface to the Second Edition

Metric Spaces is one of the foundational subjects in Modern Analysis. This second edition has been refined and expanded to provide a rigorous, clear, and comprehensive treatment of metric space theory, complete with geometric intuition, detailed proofs, and extensive examples.

List of Symbols

$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$\mathbb{N}$	Set of natural numbers
$\mathbb{Z}$	Set of integers
$\mathbb{Q}$	Set of rational numbers
$d(x, y)$	Distance between points x and y in a metric space (X, d)
$B(x, r)$	Open sphere (or ball) centered at x with radius $r > 0$
$\overline{B}(x, r)$	Closed sphere (or ball) centered at x with radius $r > 0$
$S(x, r)$	Sphere centered at x with radius $r > 0$
$\text{Int}(A)$	Interior of set A
$\overline{A}$	Closure of set A
$\text{Bd}(A)$	Boundary of set A
A'	Derived set (set of all limit points) of A
$\text{diam}(A)$	Diameter of set A
$d(A, B)$	Distance between sets A and B
ℓ_p	Space of sequences (x_n) with $\sum x_n ^p < \infty$
ℓ_∞	Space of bounded sequences
$\mathbf{C}[\mathbf{a}, \mathbf{b}]$	Space of continuous real-valued functions on $[a, b]$

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Chapter 1

Preliminaries

This chapter reviews foundational set-theoretic concepts, mappings, relations, sequences, countability, and classical real-variable inequalities that form the indispensable framework for metric space theory.

1.1 Sets

A **set** is a well-defined collection of distinct objects, called elements or members of the set. If x is an element of a set A , we write $x \in A$; otherwise, $x \notin A$.

Definition 1.1 (Set Operations). Let A and B be subsets of a universal set U .

- (i) **Union:** $A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$.
- (ii) **Intersection:** $A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$.
- (iii) **Difference:** $A \setminus B = \{x \in U : x \in A \text{ and } x \notin B\}$.
- (iv) **Complement:** $A^c = U \setminus A = \{x \in U : x \notin A\}$.
- (v) **Symmetric Difference:** $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

Theorem 1.2 (De Morgan's Laws). For any family of sets $\{A_\alpha\}_{\alpha \in \Delta}$ in a universe U :

$$\left(\bigcup_{\alpha \in \Delta} A_\alpha \right)^c = \bigcap_{\alpha \in \Delta} A_\alpha^c, \quad \left(\bigcap_{\alpha \in \Delta} A_\alpha \right)^c = \bigcup_{\alpha \in \Delta} A_\alpha^c.$$

Definition 1.3 (Cartesian Product). The **Cartesian product** of two sets A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

1.2 Functions

Definition 1.4 (Function). Let X and Y be non-empty sets. A **function** (or **mapping**) f from X into Y , written $f : X \rightarrow Y$, is a rule that assigns to each element $x \in X$ a unique element $f(x) \in Y$. The set X is called the **domain** of f , and Y is called the **codomain** of f . The **range** (or image) of f is $f(X) = \{f(x) \in Y : x \in X\}$.

Definition 1.5 (Properties of Functions). A function $f : X \rightarrow Y$ is said to be:

- (i) **Injective** (or one-to-one) if $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in X$.

- (ii) **Surjective** (or onto) if $f(X) = Y$, i.e., for every $y \in Y$, there exists $x \in X$ such that $f(x) = y$.
- (iii) **Bijective** if f is both injective and surjective.

Definition 1.6 (Image and Inverse Image). Let $f : X \rightarrow Y$ be a function.

- (i) If $A \subseteq X$, the **image** of A under f is $f(A) = \{f(x) : x \in A\}$.
- (ii) If $B \subseteq Y$, the **inverse image** (or preimage) of B under f is $f^{-1}(B) = \{x \in X : f(x) \in B\}$.

Theorem 1.7 (Properties of Inverse Images). Let $f : X \rightarrow Y$ be a mapping, and let $\{B_\alpha\}_{\alpha \in \Delta}$ be a collection of subsets of Y . Then:

$$f^{-1}\left(\bigcup_{\alpha \in \Delta} B_\alpha\right) = \bigcup_{\alpha \in \Delta} f^{-1}(B_\alpha), \quad f^{-1}\left(\bigcap_{\alpha \in \Delta} B_\alpha\right) = \bigcap_{\alpha \in \Delta} f^{-1}(B_\alpha),$$

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B) \quad \text{for any } B \subseteq Y.$$

1.3 Relations

Definition 1.8 (Equivalence Relation). A binary relation R on a set X (written $x \sim y$) is an **equivalence relation** if it satisfies:

- (i) **Reflexivity**: $x \sim x$ for all $x \in X$.
- (ii) **Symmetry**: $x \sim y \implies y \sim x$ for all $x, y \in X$.
- (iii) **Transitivity**: $x \sim y$ and $y \sim z \implies x \sim z$ for all $x, y, z \in X$.

1.4 Sequences

A **sequence** in a set X is a function $x : \mathbb{N} \rightarrow X$, usually written as $(x_n)_{n=1}^\infty$ or simply (x_n) , where $x_n = x(n)$ is the n -th term of the sequence.

1.5 Countable and Uncountable Sets

Definition 1.9 (Countability). A set A is called **finite** if it is empty or can be put in one-to-one correspondence with $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. A set A is **denumerable** (or **countably infinite**) if there exists a bijection $f : \mathbb{N} \rightarrow A$. A set is **countable** if it is either finite or denumerable; otherwise, it is **uncountable**.

Theorem 1.10. (i) Every subset of a countable set is countable.

(ii) The countable union of countable sets is countable.

(iii) The set of rational numbers $\mathbb{Q}$ is countable.

(iv) The set of real numbers $\mathbb{R}$ and the interval $[0, 1]$ are uncountable.

1.6 Inequalities

The following classical inequalities play a fundamental role in establishing metric properties on sequence and function spaces.

Theorem 1.11 (Cauchy-Schwarz Inequality). *For any real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$:*

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

with equality holding if and only if there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ (or $b_i = \lambda a_i$) for all $i = 1, \dots, n$.

Theorem 1.12 (Young's Inequality). *Let $p > 1$ and $q > 1$ be conjugate exponents, i.e., $\frac{1}{p} + \frac{1}{q} = 1$. Then for any $a \geq 0, b \geq 0$:*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality if and only if $a^p = b^q$.

Theorem 1.13 (Hölder's Inequality for Sums). *Let $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$. For any complex numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$:*

$$\sum_{i=1}^n |a_i b_i| \leq \left(\sum_{i=1}^n |a_i|^p \right)^{1/p} \left(\sum_{i=1}^n |b_i|^q \right)^{1/q}.$$

In sequence space notation, if $(a_k) \in \ell_p$ and $(b_k) \in \ell_q$:

$$\|ab\|_1 \leq \|a\|_p \|b\|_q.$$

Theorem 1.14 (Minkowski's Inequality for Sums). *Let $p \geq 1$. For any complex numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$:*

$$\left(\sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |a_i|^p \right)^{1/p} + \left(\sum_{i=1}^n |b_i|^p \right)^{1/p}.$$

This asserts that the p -norm satisfies the triangle inequality on $\mathbb{R}^n$ and on ℓ_p .

Theorem 1.15 (Hölder's and Minkowski's Inequalities for Integrals). *Let $f, g \in C[a, b]$ (or $L_p[a, b]$) with $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$.*

$$(\textbf{Hölder}) \quad \int_a^b |f(x)g(x)| dx \leq \left(\int_a^b |f(x)|^p dx \right)^{1/p} \left(\int_a^b |g(x)|^q dx \right)^{1/q}.$$

$$(\textbf{Minkowski}) \quad \left(\int_a^b |f(x) + g(x)|^p dx \right)^{1/p} \leq \left(\int_a^b |f(x)|^p dx \right)^{1/p} + \left(\int_a^b |g(x)|^p dx \right)^{1/p}.$$

Chapter 2

Introductory Concepts

This chapter introduces the central object of study: a **metric space**. We examine canonical examples, sphere geometries, open and closed sets, interior, closure, boundary, distance between sets, diameter, subspaces, and product spaces.

2.1 Definition and Examples of Metric Spaces

Definition 2.1 (Metric Space). Let X be a non-empty set. A **metric** (or distance function) on X is a real-valued function $d : X \times X \rightarrow \mathbb{R}$ satisfying the following axioms for all $x, y, z \in X$:

(M1) **Non-negativity**: $d(x, y) \geq 0$.

(M2) **Identity of Indiscernibles**: $d(x, y) = 0 \iff x = y$.

(M3) **Symmetry**: $d(x, y) = d(y, x)$.

(M4) **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a **metric space**. The elements of X are called **points**, and $d(x, y)$ is called the distance between x and y .

Example 2.2 (Real Line $\mathbb{R}$). Define $d(x, y) = |x - y|$ for all $x, y \in \mathbb{R}$. Then $(\mathbb{R}, d)$ is the usual metric space of real numbers.

Example 2.3 (Euclidean Space $\mathbb{R}^n$). For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$, define:

$$d_2(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}.$$

Then $(\mathbb{R}^n, d_2)$ is the n -dimensional Euclidean metric space.

Example 2.4 (Taxicab and Maximum Metrics on $\mathbb{R}^n$). On $\mathbb{R}^n$, we can also define:

$$\begin{aligned} \textbf{Taxicab / } \ell_1 \textbf{ Metric:} \quad d_1(x, y) &= \sum_{i=1}^n |x_i - y_i|, \\ \textbf{Maximum / } \ell_\infty \textbf{ Metric:} \quad d_\infty(x, y) &= \max_{1 \leq i \leq n} |x_i - y_i|. \end{aligned}$$

Both $(\mathbb{R}^n, d_1)$ and $(\mathbb{R}^n, d_\infty)$ are metric spaces.

Example 2.5 (Discrete Metric Space). Let X be any non-empty set. Define $d : X \times X \rightarrow \mathbb{R}$ by:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

Then d is called the **discrete metric**, and (X, d) is a discrete metric space.

Example 2.6 (Sequence Space ℓ_p ($1 \leq p < \infty$)). The set ℓ_p consists of all infinite sequences $x = (x_n)_{n=1}^{\infty}$ of real (or complex) numbers such that $\sum_{n=1}^{\infty} |x_n|^p < \infty$. The metric is defined by:

$$d_p(x, y) = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{1/p}.$$

By Minkowski's inequality, d_p satisfies the triangle inequality.

Example 2.7 (Space of Bounded Sequences ℓ_{∞}). The set ℓ_{∞} consists of all bounded sequences $x = (x_n)_{n=1}^{\infty}$ with metric:

$$d_{\infty}(x, y) = \sup_{n \geq 1} |x_n - y_n|.$$

Example 2.8 (Function Space $C[a, b]$). Let $C[a, b]$ be the set of all continuous real-valued functions defined on a closed bounded interval $[a, b]$. The uniform metric is defined by:

$$d_{\infty}(f, g) = \max_{x \in [a, b]} |f(x) - g(x)|.$$

We can also define the L_1 metric on $C[a, b]$ by:

$$d_1(f, g) = \int_a^b |f(x) - g(x)| dx.$$

2.2 Open Spheres and Closed Spheres

Definition 2.9 (Spheres and Balls). Let (X, d) be a metric space, $x_0 \in X$, and $r > 0$.

(i) The **open sphere** (or open ball) centered at x_0 with radius r is:

$$B(x_0, r) = \{x \in X : d(x, x_0) < r\}.$$

(ii) The **closed sphere** (or closed ball) centered at x_0 with radius r is:

$$\overline{B}(x_0, r) = \{x \in X : d(x, x_0) \leq r\}.$$

(iii) The **sphere** centered at x_0 with radius r is:

$$S(x_0, r) = \{x \in X : d(x, x_0) = r\}.$$

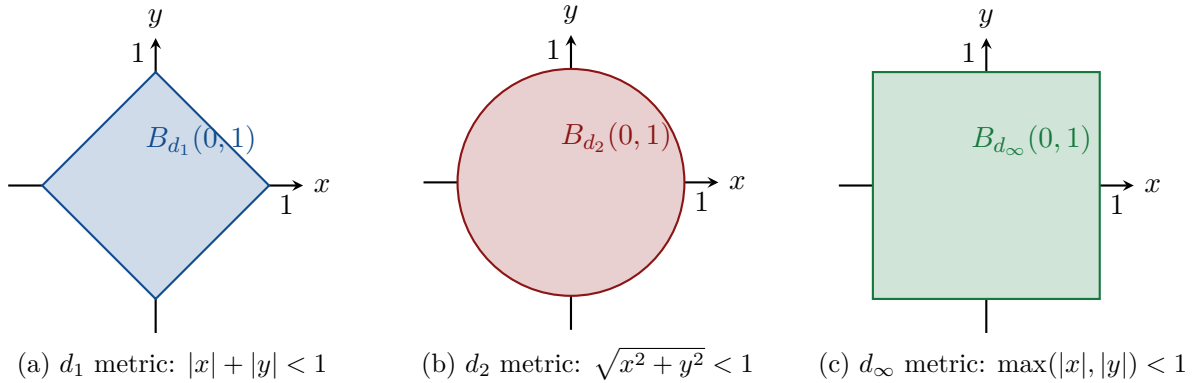


Figure 2.1: Unit open balls centered at the origin $(0, 0)$ in $\mathbb{R}^2$ under three distinct metrics.

Theorem 2.10. In any metric space (X, d) , every open sphere $B(x_0, r)$ is an open set, and every closed sphere $\overline{B}(x_0, r)$ is a closed set.

2.3 Neighbourhoods and Open Sets

Definition 2.11 (Neighbourhood). Let (X, d) be a metric space and $x \in X$. A subset $N \subseteq X$ is called a **neighbourhood** of x if there exists $r > 0$ such that $B(x, r) \subseteq N$.

Definition 2.12 (Open Set). A subset $G \subseteq X$ is called an **open set** if G is a neighbourhood of each of its points, i.e., for every $x \in G$, there exists $r_x > 0$ such that $B(x, r_x) \subseteq G$.

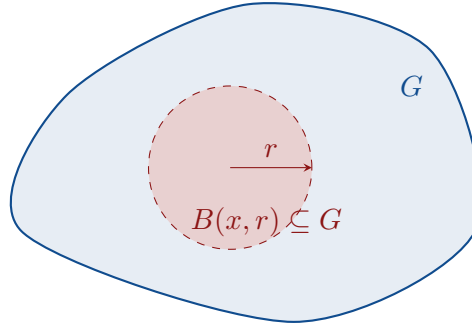


Figure 2.2: Geometric characterisation of an open set G : for any $x \in G$, an open sphere $B(x, r)$ is completely contained in G .

Theorem 2.13 (Topology of Open Sets). *In any metric space (X, d) :*

- (i) *The empty set $\emptyset$ and the whole space X are open sets.*
- (ii) *The union of an arbitrary family of open sets is open.*
- (iii) *The intersection of a finite family of open sets is open.*

2.4 Interior Points, Closed Sets, Limit Points, and Closure

Definition 2.14 (Interior Point and Interior). Let $A \subseteq X$. A point $x \in A$ is an **interior point** of A if there exists $r > 0$ such that $B(x, r) \subseteq A$. The set of all interior points of A is called the **interior** of A , denoted by $\text{Int}(A)$ or A° .

Theorem 2.15. *For any $A \subseteq X$, $\text{Int}(A)$ is the largest open set contained in A . Furthermore, A is open if and only if $\text{Int}(A) = A$.*

Definition 2.16 (Closed Set). A subset $F \subseteq X$ is called a **closed set** if its complement $F^c = X \setminus F$ is an open set.

Definition 2.17 (Limit Point and Derived Set). A point $x \in X$ is called a **limit point** (or accumulation point) of $A \subseteq X$ if every open sphere $B(x, r)$ contains at least one point of A other than x , i.e.,

$$(B(x, r) \setminus \{x\}) \cap A \neq \emptyset \quad \forall r > 0.$$

The set of all limit points of A is called the **derived set** of A , denoted by A' .

Theorem 2.18. *A set $F \subseteq X$ is closed if and only if it contains all its limit points, i.e., $A' \subseteq F$.*

Definition 2.19 (Closure of a Set). The **closure** of $A \subseteq X$, denoted by $\overline{A}$ or $\text{Cl}(A)$, is defined as $\overline{A} = A \cup A'$.

Theorem 2.20. *The closure $\overline{A}$ is the smallest closed set containing A . Moreover, A is closed if and only if $\overline{A} = A$.*

Definition 2.21 (Boundary Points). A point $x \in X$ is a **boundary point** of $A \subseteq X$ if every open sphere $B(x, r)$ contains at least one point of A and at least one point of $X \setminus A$. The set of all boundary points of A is the **boundary** $\text{Bd}(A) = \overline{A} \cap \overline{X \setminus A}$.

2.5 Distance Between Sets, Subspaces, and Product Spaces

Definition 2.22 (Distance and Diameter). Let (X, d) be a metric space, $A, B \subseteq X$, and $x \in X$.

- (i) Distance from x to A : $d(x, A) = \inf\{d(x, a) : a \in A\}$.
- (ii) Distance between A and B : $d(A, B) = \inf\{d(a, b) : a \in A, b \in B\}$.
- (iii) Diameter of A : $\text{diam}(A) = \sup\{d(a_1, a_2) : a_1, a_2 \in A\}$.

Definition 2.23 (Subspace Metric). Let (X, d) be a metric space and $Y \subseteq X$ a non-empty subset. The restriction $d_Y = d|_{Y \times Y}$ is a metric on Y , and (Y, d_Y) is called a **metric subspace** of (X, d) .

Chapter 3

Completeness

Completeness is a foundational concept in mathematical analysis that guarantees sequence convergence within a space without missing limit points. This chapter investigates Cauchy sequences, complete metric spaces, density, separability, Baire's Category Theorem, and metric space completion.

3.1 Convergent Sequences and Cauchy Sequences

Definition 3.1 (Convergent Sequence). A sequence (x_n) in a metric space (X, d) is said to **converge** to a point $x \in X$, written $x_n \rightarrow x$ or $\lim_{n \rightarrow \infty} x_n = x$, if for every $\varepsilon > 0$, there exists an integer $N \in \mathbb{N}$ such that:

$$d(x_n, x) < \varepsilon \quad \text{for all } n \geq N.$$

Theorem 3.2 (Uniqueness and Boundedness). *In any metric space (X, d) :*

- (i) *Every convergent sequence has a unique limit.*
- (ii) *Every convergent sequence is bounded.*

Definition 3.3 (Cauchy Sequence). A sequence (x_n) in a metric space (X, d) is called a **Cauchy sequence** if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that:

$$d(x_m, x_n) < \varepsilon \quad \text{for all } m, n \geq N.$$

Theorem 3.4. *Every convergent sequence in a metric space is a Cauchy sequence.*

3.2 Complete Metric Spaces

Definition 3.5 (Complete Metric Space). A metric space (X, d) is called **complete** if every Cauchy sequence in X converges to a point in X .

Theorem 3.6 (Completeness of Canonical Spaces). (i) *The real line $(\mathbb{R}, |\cdot|)$ and complex plane $(\mathbb{C}, |\cdot|)$ are complete metric spaces.*

- (ii) *Euclidean space $(\mathbb{R}^n, d_2)$ is complete.*
- (iii) *The sequence space ℓ_p ($1 \leq p < \infty$) is complete.*
- (iv) *The space of bounded sequences ℓ_∞ is complete.*
- (v) *The function space $C[a, b]$ under the uniform metric $d_\infty(f, g) = \max_{x \in [a, b]} |f(x) - g(x)|$ is complete.*

Example 3.7 (Incomplete Metric Space). The rational numbers $(\mathbb{Q}, |\cdot|)$ is incomplete because sequences of rationals approximating $\sqrt{2}$ are Cauchy in $\mathbb{Q}$ but fail to converge in $\mathbb{Q}$.

Theorem 3.8 (Closed Subsets of Complete Spaces). *Let (X, d) be a complete metric space and $Y \subseteq X$. The subspace $(Y, d|_Y)$ is complete if and only if Y is a closed subset of X .*

3.3 Dense Sets, Separability, and Nowhere Dense Sets

Definition 3.9 (Dense Set). A subset A of a metric space (X, d) is called **dense** in X if $\overline{A} = X$, i.e., for every $x \in X$ and every $\varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$.

Definition 3.10 (Separable Space). A metric space (X, d) is called **separable** if it contains a countable dense subset.

Theorem 3.11 (Separability Examples). (i) $\mathbb{R}^n$ is separable because $\mathbb{Q}^n$ is a countable dense subset.

(ii) ℓ_p ($1 \leq p < \infty$) is separable.

(iii) ℓ_∞ is **not** separable.

(iv) $C[a, b]$ is separable (by the Weierstrass Approximation Theorem using polynomials with rational coefficients).

Definition 3.12 (Nowhere Dense Set). A subset $A \subseteq X$ is called **nowhere dense** in X if $\text{Int}(\overline{A}) = \emptyset$, i.e., the closure of A contains no open balls.

3.4 Baire's Category Theorem

Definition 3.13 (Meagre / First Category Sets). A subset $A \subseteq X$ is of **first category** (or **meagre**) in X if $A = \bigcup_{n=1}^{\infty} A_n$, where each A_n is nowhere dense in X . A set that is not of first category is of **second category** (or **non-meagre**).

Theorem 3.14 (Baire's Category Theorem). *Every complete metric space (X, d) is of second category in itself. Equivalent statements:*

(i) *If (X, d) is complete and $X = \bigcup_{n=1}^{\infty} F_n$ where each F_n is closed, then at least one F_n has a non-empty interior ($\text{Int}(F_n) \neq \emptyset$).*

(ii) *The intersection of any countable collection of dense open sets in a complete metric space is dense.*

3.5 Completion of a Metric Space

Theorem 3.15 (Metric Space Completion). *For every metric space (X, d) , there exists a complete metric space $(\widehat{X}, \widehat{d})$ and an isometry $W : X \rightarrow \widehat{X}$ such that $W(X)$ is dense in $\widehat{X}$. Furthermore, $(\widehat{X}, \widehat{d})$ is unique up to isometry.*

Chapter 4

Continuous Functions

Continuity is the central structural morphism between metric spaces. This chapter provides foundational characterisations of continuous maps, uniform continuity, homeomorphisms, and extension theorems.

4.1 Definition and Characterisations of Continuity

Definition 4.1 (Continuous Function at a Point). Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \rightarrow Y$ is called **continuous at a point** $x_0 \in X$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that:

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

In ball notation, $f(B_X(x_0, \delta)) \subseteq B_Y(f(x_0), \varepsilon)$. A function f is **continuous on** X if it is continuous at every point $x \in X$.

Theorem 4.2 (Topological and Sequential Characterisations of Continuity). *Let $f : (X, d_X) \rightarrow (Y, d_Y)$ be a mapping. The following statements are equivalent:*

- (i) f is continuous on X .
- (ii) For every open set $V \subseteq Y$, the inverse image $f^{-1}(V)$ is an open set in X .
- (iii) For every closed set $F \subseteq Y$, the inverse image $f^{-1}(F)$ is a closed set in X .
- (iv) For every sequence (x_n) in X converging to x , the sequence $(f(x_n))$ converges to $f(x)$ in Y .
- (v) For every subset $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$.

4.2 Uniform Continuity and Lipschitz Continuity

Definition 4.3 (Uniform Continuity). A function $f : (X, d_X) \rightarrow (Y, d_Y)$ is **uniformly continuous** on X if for every $\varepsilon > 0$, there exists $\delta > 0$ (depending only on ε , independent of x) such that:

$$d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \varepsilon \quad \forall x_1, x_2 \in X.$$

Definition 4.4 (Lipschitz Continuity). A function $f : (X, d_X) \rightarrow (Y, d_Y)$ is called **Lipschitz continuous** if there exists a constant $K \geq 0$ such that:

$$d_Y(f(x_1), f(x_2)) \leq K d_X(x_1, x_2) \quad \forall x_1, x_2 \in X.$$

If $K < 1$, f is called a **contraction mapping**.

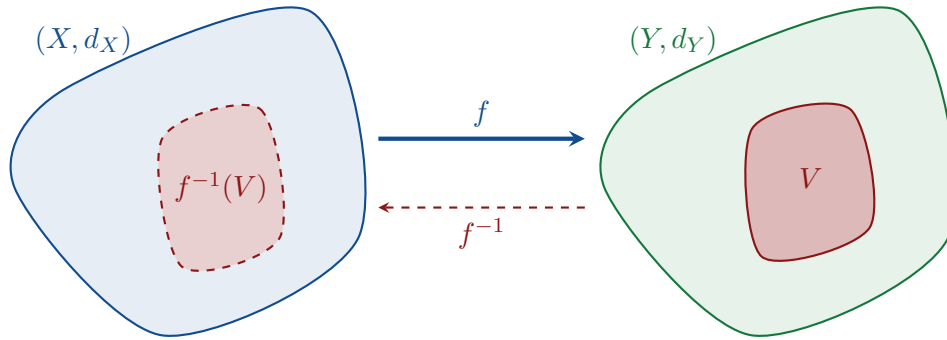


Figure 4.1: Continuous mapping $f : X \rightarrow Y$: the inverse image $f^{-1}(V)$ of any open set $V \subseteq Y$ is an open set in X .

4.3 Homeomorphisms and Equivalent Metrics

Definition 4.5 (Homeomorphism). A function $f : (X, d_X) \rightarrow (Y, d_Y)$ is called a **homeomorphism** if:

- (i) f is bijective (one-to-one and onto).
- (ii) f is continuous.
- (iii) f^{-1} is continuous.

If a homeomorphism exists between X and Y , the spaces are said to be **homeomorphic** ($X \cong Y$).

Definition 4.6 (Uniformly Equivalent Metrics). Two metrics d_1 and d_2 on the same set X are called **equivalent** if they generate the same open sets, and **uniformly equivalent** if the identity map $i : (X, d_1) \rightarrow (X, d_2)$ is a uniform homeomorphism.

Chapter 5

Compactness

Compactness generalizes finite set properties to infinite spaces, playing a central role in real analysis, topology, and optimization. This chapter examines open covers, sequential compactness, total boundedness, finite intersection property, and continuous maps on compact domains.

5.1 Compact Spaces and Sets

Definition 5.1 (Open Cover and Subcover). Let (X, d) be a metric space and $A \subseteq X$.

- (i) A collection $\mathcal{U} = \{G_\alpha\}_{\alpha \in \Delta}$ of open sets in X is an **open cover** of A if $A \subseteq \bigcup_{\alpha \in \Delta} G_\alpha$.
- (ii) A subcollection $\mathcal{V} \subseteq \mathcal{U}$ is a **subcover** of A if $A \subseteq \bigcup_{V \in \mathcal{V}} V$.

Definition 5.2 (Compact Metric Space). A metric space (X, d) (or a subset $K \subseteq X$) is called **compact** if every open cover of X (or K) admits a finite subcover.

Theorem 5.3 (Heine-Borel Theorem in $\mathbb{R}^n$). *A subset $K \subseteq \mathbb{R}^n$ (with the Euclidean metric d_2) is compact if and only if it is closed and bounded.*

Theorem 5.4 (Properties of Compact Sets). (i) *Every compact subset of a metric space is closed and bounded.*

- (ii) *Every closed subset of a compact metric space is compact.*

5.2 Sequential Compactness and Total Boundedness

Definition 5.5 (Sequential Compactness). A metric space (X, d) is called **sequentially compact** if every sequence (x_n) in X has a convergent subsequence whose limit lies in X .

Definition 5.6 (ε -Net and Total Boundedness). Let (X, d) be a metric space and $\varepsilon > 0$.

- (i) A subset $E \subseteq X$ is called an **ε -net** for X if $X = \bigcup_{x \in E} B(x, \varepsilon)$.
- (ii) X is called **totally bounded** if for every $\varepsilon > 0$, there exists a finite ε -net for X .

Theorem 5.7 (Equivalence of Compactness Notions). *In any metric space (X, d) , the following properties are equivalent:*

- (i) *X is compact.*
- (ii) *X is sequentially compact.*
- (iii) *X is complete and totally bounded.*

5.3 Finite Intersection Property

Definition 5.8 (Finite Intersection Property). A collection $\mathcal{F}$ of subsets of X has the **Finite Intersection Property (FIP)** if every finite subcollection $\{F_1, F_2, \dots, F_k\} \subseteq \mathcal{F}$ has a non-empty intersection:

$$\bigcap_{i=1}^k F_i \neq \emptyset.$$

Theorem 5.9 (Compactness via FIP). *A metric space (X, d) is compact if and only if every collection $\mathcal{F}$ of closed subsets of X with the Finite Intersection Property has a non-empty intersection:*

$$\bigcap_{F \in \mathcal{F}} F \neq \emptyset.$$

5.4 Continuous Functions on Compact Spaces

Theorem 5.10 (Preservation of Compactness). *Let $f : (X, d_X) \rightarrow (Y, d_Y)$ be a continuous mapping. If $K \subseteq X$ is compact, then its image $f(K)$ is compact in Y .*

Theorem 5.11 (Extreme Value Theorem). *Let (X, d) be a compact metric space and $f : X \rightarrow \mathbb{R}$ a continuous real-valued function. Then f is bounded and attains its maximum and minimum values on X , i.e., there exist $x_m, x_M \in X$ such that:*

$$f(x_m) \leq f(x) \leq f(x_M) \quad \forall x \in X.$$

Theorem 5.12 (Heine-Cantor Theorem). *If (X, d_X) is a compact metric space, every continuous function $f : X \rightarrow Y$ is uniformly continuous on X .*

Chapter 6

Connectedness

Connectedness formalizes the notion of a space consisting of a single "piece". This chapter explores separated sets, connected spaces, connected components, connected subsets of $\mathbb{R}$, and the generalized Intermediate Value Theorem.

6.1 Separated Sets and Connected Spaces

Definition 6.1 (Separated Sets). Two non-empty subsets A and B of a metric space (X, d) are called **separated** if:

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Definition 6.2 (Connected Space). A metric space (X, d) is called **disconnected** if X can be written as the union of two non-empty separated sets A and B , i.e., $X = A \cup B$ with $A \neq \emptyset, B \neq \emptyset$, and $\overline{A} \cap B = \emptyset = A \cap \overline{B}$. A space X is **connected** if it is not disconnected.

Theorem 6.3 (Equivalent Characterisations of Connectedness). *For any metric space (X, d) , the following statements are equivalent:*

- (i) X is connected.
- (ii) The only subsets of X that are both open and closed (clopen) are $\emptyset$ and X .
- (iii) X cannot be written as the union of two disjoint non-empty open sets.
- (iv) X cannot be written as the union of two disjoint non-empty closed sets.
- (v) Every continuous function $f : X \rightarrow \{0, 1\}$ (with discrete metric on $\{0, 1\}$) is constant.

6.2 Connected Components

Definition 6.4 (Connected Component). Let (X, d) be a metric space and $x \in X$. The **connected component** of x , denoted $C(x)$, is the union of all connected subsets of X containing x :

$$C(x) = \bigcup \{E \subseteq X : x \in E \text{ and } E \text{ is connected}\}.$$

Theorem 6.5 (Properties of Components). (i) *For each $x \in X$, $C(x)$ is a maximal connected subset of X .*

- (ii) $C(x)$ is a closed set in X .
- (iii) *The connected components form a partition of X into pairwise disjoint closed connected sets.*

6.3 Connected Subsets of $\mathbb{R}$ and Continuous Images

Theorem 6.6 (Connected Subsets of $\mathbb{R}$). *A subset $E \subseteq \mathbb{R}$ (with the standard metric) is connected if and only if E is an interval (open, closed, half-open, or infinite).*

Theorem 6.7 (Preservation of Connectedness). *Let $f : (X, d_X) \rightarrow (Y, d_Y)$ be a continuous mapping. If $E \subseteq X$ is connected, then its image $f(E)$ is connected in Y .*

Theorem 6.8 (Generalized Intermediate Value Theorem). *Let (X, d) be a connected metric space and $f : X \rightarrow \mathbb{R}$ a continuous function. If $a, b \in X$ and $c \in \mathbb{R}$ is any value between $f(a)$ and $f(b)$, then there exists $x_0 \in X$ such that $f(x_0) = c$.*

Chapter 7

Fixed Point Theorems and Their Applications

Fixed point theory provides powerful tools for establishing the existence and uniqueness of solutions to algebraic, differential, and integral equations. This chapter focuses on the Banach Contraction Principle and its direct applications.

7.1 Banach Contraction Principle

Definition 7.1 (Fixed Point). Let $f : X \rightarrow X$ be a mapping on a set X . A point $x^* \in X$ is called a **fixed point** of f if $f(x^*) = x^*$.

Definition 7.2 (Contraction Mapping). Let (X, d) be a metric space. A mapping $f : X \rightarrow X$ is called a **contraction mapping** (or contraction) if there exists a real constant $k \in [0, 1)$ such that:

$$d(f(x), f(y)) \leq k d(x, y) \quad \forall x, y \in X.$$

The constant k is called the contraction constant.

Theorem 7.3 (Banach Fixed Point Theorem). *Let (X, d) be a non-empty complete metric space, and let $f : X \rightarrow X$ be a contraction mapping with constant $k < 1$. Then:*

- (i) f has a unique fixed point $x^* \in X$ (i.e., $f(x^*) = x^*$).
- (ii) For any arbitrary initial point $x_0 \in X$, the sequence of Picard iterations defined by $x_{n+1} = f(x_n)$ ($n = 0, 1, 2, \dots$) converges to x^* :

$$\lim_{n \rightarrow \infty} x_n = x^*.$$

- (iii) The following error bounds hold for all $n \geq 1$:

$$\textbf{Prior Error Estimate:} \quad d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_0, x_1),$$

$$\textbf{Posterior Error Estimate:} \quad d(x_n, x^*) \leq \frac{k}{1-k} d(x_{n-1}, x_n).$$

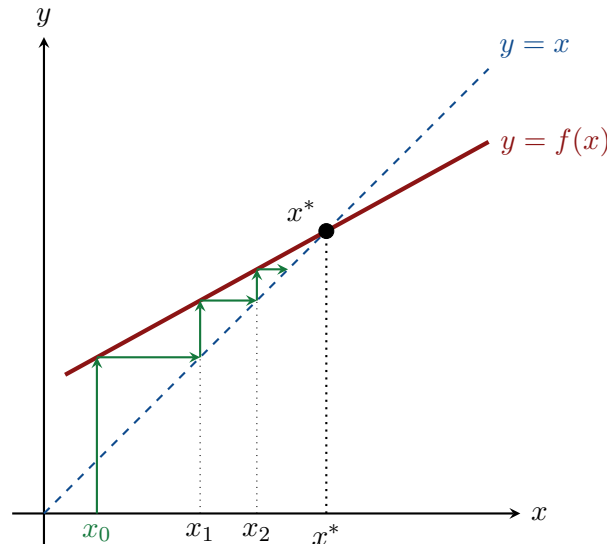


Figure 7.1: Picard cobweb iteration $x_{n+1} = f(x_n)$ converging rapidly to the unique fixed point x^* of contraction f .

7.2 Application to Integral Equations

Theorem 7.4 (Fredholm Integral Equations). *Consider the Fredholm integral equation of the second kind:*

$$y(x) = g(x) + \lambda \int_a^b K(x, t) y(t) dt, \quad x \in [a, b],$$

where $g \in C[a, b]$, the kernel $K \in C([a, b] \times [a, b])$, and $\lambda \in \mathbb{R}$. Let $M = \max_{(x,t) \in [a,b] \times [a,b]} |K(x, t)|$. If $|\lambda| < \frac{1}{M(b-a)}$, then the integral equation has a unique continuous solution $y \in C[a, b]$.

Theorem 7.5 (Volterra Integral Equations). *Consider the Volterra integral equation of the second kind:*

$$y(x) = g(x) + \lambda \int_a^x K(x, t) y(t) dt, \quad x \in [a, b].$$

For any continuous kernel K and continuous function g , the Volterra equation has a unique continuous solution $y \in C[a, b]$ for **every** parameter $\lambda \in \mathbb{R}$.

7.3 Application to Inverse and Implicit Function Theorems

The Banach Contraction Principle can be used to construct iterative proofs of:

- (i) **Picard-Lindelöf Existence Theorem:** For initial value problems $y' = f(x, y)$, $y(x_0) = y_0$ with Lipschitz continuous f .
- (ii) **Inverse Function Theorem:** Guaranteeing local C^1 invertibility of $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ when $Df(x_0)$ is non-singular.
- (iii) **Implicit Function Theorem:** Guaranteeing existence of unique explicit solution $y = g(x)$ for system $F(x, y) = 0$.

Appendix A

The Cantor Set

The Cantor ternary set is a classical counterexample in analysis and topology, illustrating a set that is uncountable, compact, nowhere dense, and has measure zero.

A.1 Construction of the Cantor Set

Start with the closed unit interval $C_0 = [0, 1]$.

1. Remove the open middle third $(1/3, 2/3)$, leaving $C_1 = [0, 1/3] \cup [2/3, 1]$.
2. Remove the open middle third of each of the two remaining intervals, leaving $C_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$.
3. Repeat this process indefinitely. At step n , C_n consists of 2^n closed intervals, each of length 3^{-n} .

The **Cantor set** C is the intersection of all C_n :

$$C = \bigcap_{n=0}^{\infty} C_n.$$

A.2 Properties of the Cantor Set

Theorem A.1. *The Cantor set C satisfies:*

- (i) C is compact (closed and bounded in $\mathbb{R}$).
- (ii) C is nowhere dense in $\mathbb{R}$ ($\text{Int}(C) = \emptyset$).
- (iii) C is perfect (it has no isolated points, $C' = C$).
- (iv) C is uncountable ($\text{card}(C) = c = 2^{\aleph_0}$, via ternary expansion $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$, $a_n \in \{0, 2\}$).
- (v) C has Lebesgue measure zero ($\lambda(C) = 1 - \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = 0$).

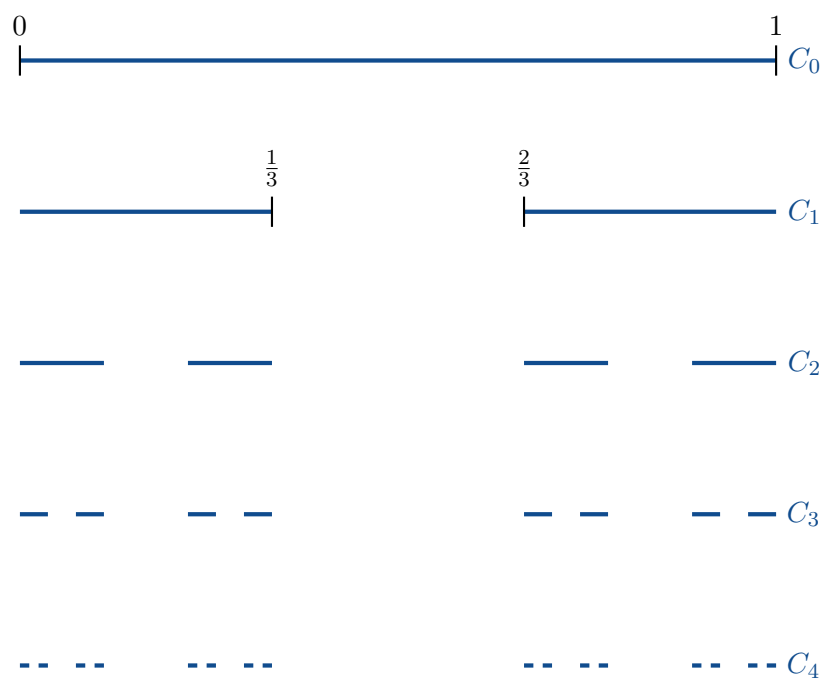


Figure A.1: Step-by-step construction of the Cantor ternary set $C = \bigcap_{n=0}^{\infty} C_n$.

Appendix B

Limits in Metric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, $f : A \rightarrow Y$, and $x_0 \in A'$.

Definition B.1 (Limit of a Function). We say that $L \in Y$ is the **limit of $f(x)$ as x approaches x_0** , written $\lim_{x \rightarrow x_0} f(x) = L$, if for every $\varepsilon > 0$, there exists $\delta > 0$ such that:

$$0 < d_X(x, x_0) < \delta \implies d_Y(f(x), L) < \varepsilon.$$

Theorem B.2 (Sequential Characterisation of Limits). $\lim_{x \rightarrow x_0} f(x) = L$ if and only if for every sequence (x_n) in $A \setminus \{x_0\}$ with $x_n \rightarrow x_0$, $f(x_n) \rightarrow L$.